

Algebra 1
Unit 5
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Use the system of equations $\begin{cases} x + 4y = 3 \\ 0.5x - 4y = 6 \end{cases}$ to complete questions 1 – 4.

1. Complete the statement.

- ☒ Adding
☐ Subtracting

the original equations in this system would result in eliminating one of the variables,

- ☐ x .
☒ y .

2. Perform the selected operation selected in question 1.

$$\begin{array}{r} x + 4y = 3 \\ +0.5x - 4y = 6 \\ \hline 1.5x = 9 \end{array}$$

3. Solve the system using your equation from question 2.

$$1.5x = 9$$

$$x = 6$$

$$x + 4y = 3$$

$$6 + 4y = 3$$

$$4y = -3$$

$$y = -\frac{3}{4}$$

$$\left(6, -\frac{3}{4}\right)$$

or

$$(6, -0.75)$$

4. Verify the solution algebraically using substitution.

$$6 + 4\left(-\frac{3}{4}\right) = 6 - 3 = 3 \checkmark$$

$$0.5(6) - 4\left(-\frac{3}{4}\right) = 3 + 3 = 6 \checkmark$$

5. Solve the system of equations using elimination.

$$\begin{cases} 5p + 2h = 13 \\ (3p - h = -12)2 \end{cases}$$

$$\begin{array}{r} 5p + 2h = 13 \\ +6p - 2h = -24 \\ \hline 11p = -11 \\ p = -1 \end{array}$$

$$\begin{array}{r} 5(-1) + 2h = 13 \\ -5 + 2h = 13 \end{array}$$

$$2h = 18$$

$$h = 9$$

$$(-1, 9)$$

6. Verify the solution from question 5 algebraically using substitution.

$$5(-1) + 2(9) = -5 + 18 = 13 \checkmark$$

$$3(-1) - 9 = -3 - 9 = -12 \checkmark$$

7. Solve the system of equations algebraically.

$$\begin{array}{lcl}
 \begin{cases} 3x + y = 5 \\ 5x - 4y = -12 \end{cases} & \begin{array}{l} 12x + 4y = 20 \\ +5x - 4y = -12 \\ \hline 17x = 8 \\ x = \frac{8}{17} \end{array} & \begin{array}{l} 3\left(\frac{8}{17}\right) + y = 5 \\ \frac{24}{17} + y = 5 \\ y = \frac{85}{17} - \frac{24}{17} \\ y = \frac{61}{17} \end{array}
 \end{array}
 \quad \left(\frac{8}{17}, \frac{61}{17}\right)$$

8. Verify the solution from question 7 algebraically using substitution.

$$\begin{aligned}
 3\left(\frac{8}{17}\right) + \left(\frac{61}{17}\right) &= \frac{24}{17} + \frac{61}{17} = \frac{85}{17} = 5 \checkmark \\
 5\left(\frac{8}{17}\right) - 4\left(\frac{61}{17}\right) &= \frac{40}{17} - \frac{244}{17} = -\frac{204}{17} = -12 \checkmark
 \end{aligned}$$

9. Mia is solving the system of equations $\begin{cases} 2x + 3y = 7 \\ 3x - y = -6 \end{cases}$ using elimination. In order to eliminate the x , she multiplies the first equation by a factor of 3 and the second equation by a factor of 2. Explain whether her method will work.

Yes, the x terms will be eliminated by subtracting the equations.

$$\begin{aligned}
 3(2x + 3y = 7) &\rightarrow 6x + 9y = 21 \\
 2(3x - y = -6) &\rightarrow 6x - 2y = -12
 \end{aligned}$$

10. Mia's system would be simpler to solve by eliminating the y . Describe why this is true and then solve Mia's system.

Only the second equation needs to be manipulated to eliminate the y .

$$3(3x - y = -6) \rightarrow 9x - 3y = -18$$

$$\begin{array}{lcl}
 \begin{array}{l} 2x + 3y = 7 \\ +9x - 3y = -18 \\ \hline 11x = -11 \\ x = -1 \end{array} & \begin{array}{l} 2(-1) + 3y = 7 \\ -2 + 3y = 7 \\ 3y = 9 \\ y = 3 \end{array} & (-1, 3)
 \end{array}$$